

Peter Little

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Education

University of California, San Diego

B.S. Data Science

Relevant coursework: Data Structures & Algorithms (DSC 30), Data Science Fundamentals (DSC 10 & 20), Theoretical Foundations of Data Science (DSC 40A)

La Jolla, CA
Expected June 2028

Experience

Triton Robotics - Embedded Team Member

- Collaborate with interdisciplinary teams to design and develop robotic systems for competition and research projects.
- Contribute to debugging and system optimization for embedded real-time data handling

La Jolla, CA
Winter 2025-Present

Robotics Project Leader

- Led a team of students in designing and building a robot for various competitions; achieved **1st Place** in local robotics contest.
- Managed project workflow, assigning coding and hardware responsibilities

Rancho Cucamonga,
CA
Spring 2024

Projects

EMG Prosthetic – Triton NeuroTech | Python, Pandas, Pytorch, Scikit Learn

- Built a **bidirectional LSTM (BiLSTM)** neural network in PyTorch to classify hand gestures from multichannel EMG biosignals, achieving **82% gesture classification accuracy**.
- Developed a **signal-processing** pipeline (60 Hz notch + bandpass filtering) to remove noise and improve EMG feature quality.
- Implemented **sliding-window time-series segmentation** to convert continuous EMG data into labeled training sequences for supervised learning.
- Benchmarked model performance across **multiple EMG datasets (MindRove + UCI)** to evaluate generalization using confusion matrices, and classification reports.

La Jolla, CA
Fall 2025 - Present

Omni-Drive Chassis Project – Triton Robotics | C++, IMU Web, CAN, Power Control

- Developed a **real-time motor control system** using CAN-based networking and IMU sensor data.
- Implemented a feedback loop for smooth robotic movement and “Beyblade Mode” spin control for stability.
- Collaborated with electrical and mechanical teams to integrate embedded control with system telemetry.

La Jolla, CA
Winter 2024 -Present

Disney Doppelgängers – DataHacks 2025 | Web scraping, BeautifulSoup, spaCy, BART

- Built an **NLP-based web app** matching users to Disney characters using text embeddings and cosine similarity.
- Scraped and processed 50+ character pages using BeautifulSoup for dataset creation.
- Applied spaCy for adjective extraction and BART for text summarization to refine personality descriptors.
- Generated sentence embeddings via **Hugging Face models** and deployed similarity search for character matching

La Jolla, CA
Spring 2025

Technical Skills

Languages: Python, Java, C++, MATLAB

Libraries/Tools: NumPy, Pandas, SciPy, scikit-learn, Pytorch, Torchvision, spaCy, Hugging Face, BeautifulSoup, Matplotlib, Git, VSCode, IntelliJ, Arduino

Volunteer Experience: Triton Robotics Embedded Team Member, Triton Neurotech, CSES Innovate Research Team